



Ontology Engineering Knowledge and Reasoning

Dewi Wardani



- The flows is **similar to software engineering**
- Concerned with the challenges of designing ontologies by providing methodologies and auxiliary tools for their development, evaluation, and maintenance
- Or called **Ontology design**
- The design of large-size and complex knowledge bases requires a similarly structured approach



- requirement analysis,
- ontology creation,
 - **orthogonal issue** of ontology modularization which is particularly important for **ontology reuse** and
 - collaborative ontology creation as well as for optimizing automated inference
- ontology quality assurance



Requirement Analysis

- In the very first place, it should be decided **whether a semantic representation is at all needed or whether an alternative approach** (like using a classical database) would be a better choice
- There are essentially **two major points in favor of an ontology-based system**:
 - First, the knowledge represented **in a semantic format can be more easily exchanged** as well as integrated with knowledge from other sources.
 - Second, by employing deduction algorithms, **the implicit knowledge following from a semantic specification becomes accessible**.



- If large amounts of data have to be handled and **a less expressive formalism is sufficient**, RDF(S) might be the right choice
- If the size of the represented **information is moderate and more expressive means of knowledge representation** (as well as elaborate support for inferring implicit knowledge) are desired, **OWL DL** would be a better recommendation
- In the case of OWL 2, some profiles might also be adequate for scenarios situated in the middle of this spectrum



- Questions:
 - What **domain** has to be modeled? What aspects of this domain have to be captured?
 - What is the needed granularity, i.e. **the level of detail**, of the specification?
 - What are **the tasks to be accomplished** with the help of the ontology: browsing a body of knowledge, search for information, querying or checking inferences? What kind of inferences are expected or desired?



Ontology Creation

- Creating an ontology can be seen as **transferring knowledge into a computer-accessible form**.
- The current representation of the provided knowledge can be exploited **to facilitate the formalization process** (i.e. the process of making the inherent semantics formally explicit).
- Based on that criterion, we distinguish **human, unstructured, semi-structured, and structured sources** of knowledge which will be treated in the following sections.



- Formalizing a piece of knowledge **is more or less straightforward**, provided the underlying formalism is expressive enough
- Source
 - It's in Your Heads: Human Sources
 - It's in Your Books: Unstructured sources
 - It's on the Web: Semistructured Sources
 - It's in the Databases: Structured Sources



Quality Assurance

- Question:
 - does it **fulfill the intended purpose**? Is it possible to infer
 - does the information provided by the ontology together with a reasoning framework **help the user in accomplishing his task**?



- What Makes an Ontology Good
 - Logical Criteria
 - Structural and Formal Criteria
 - Accuracy Criteria



- How to (Not) Model Correctly
 - Don't forget **disjointness**
 - Don't forget **role characteristics**
 - Don't choose **too specific domains or ranges**
 - Be careful with **quantifiers**
 - Don't mistake parts for subclasses
 - Watch the **direction of roles**
 - Don't confuse class subsumption and class equivalence
 - Don't translate too verbally



Modular Ontologies

- The field of ontology modularization investigates how large ontologies can be composed of smaller parts, called modules.
- OWL and OWL 2 provide basic support for the distribution resp. Module aspects through **owl:imports** allowing for the inclusion of other ontologies that might be situated elsewhere on the Web



Building Ontologies

- In practical terms, developing an ontology includes:
 - **defining classes** in the ontology,
 - **arranging the classes in a taxonomic** (subclass–superclass) hierarchy,
 - **defining slots** and describing allowed values for these slots,
 - **filling in the values** for slots for instances.



Building Ontologies

- There is **no one correct way to model a domain**— there are always viable alternatives. The best solution almost always **depends on the application that you have in mind and the extensions that you anticipate.**
- Ontology development is necessarily **an iterative process.**
- **Concepts in the ontology should be close to objects (physical or logical) and relationships in your domain of interest.** These are most likely to be nouns (objects) or verbs (relationships) in sentences that describe your domain.



Step 1. Determine the domain and scope of the ontology

- That is, answer several basic questions:
 - What is **the domain** that the ontology will cover?
 - For what we are going **to use** the ontology?
 - For what **types of questions** the information in the ontology should provide answers?
 - Who will **use and maintain** the ontology?



- **Competency questions:** questions to test the scope of domain.

Ex:

- Which plant characteristics should I consider when choosing an ethnomedicine?
- Which part of kapulaga more useful, the leaves or the root ?
- Does ginger go well with lime?
- What is the best choice of ethnomedicine for pregnant women?
- Which characteristics of a ethnomedicine affect its appropriateness for fever?
- Where and when can we find Buah Merah?



Step 2. Consider reusing existing ontologies

- Sure, if they exist!!!
 - Claims ontologies made in this paper about being able import existing ontologies
- Therefore, the concept of sharing knowledge is real.
We can build an ontology by using other ontologies
 - Ex: Sebelas Maret Ontology as all knowledge about this University is built uses shared from several existing ontologies



Step 3. Enumerate important terms in the ontology

- What are the **terms** we would like to talk about?
- What **properties** do those terms have?
- What **would we like to say** about those terms?. How far the description of the terms.

Example – plants, location, a part of plants, flavor, color and ingredients; different types of ginger; subtypes of coconuts



Step 4. Define the classes and the class hierarchy

- Methods:
 - **Top-down**: define the big picture then next to the detail
 - **Bottom-up**: define specific class first and the work your way up
 - **Combination**.



- Result
 - **hierarchical** arrangements of concepts
 - If a class **A** is a superclass of class **B**, then every instance of B is also an instance of A
 - This implies “In other words, the class B represents a concept that is a **“kind of”** A.”



Step 5. Define the properties of classes—slots

DIKE
Research
Group

- Types of properties
 - “intrinsic” properties such as the color of a plant;
 - “extrinsic” properties such as a plant’s name, and area it comes from; parts, if the object is structured; these can be both physical and abstract “parts” (e.g., the courses of a meal)
 - relationships to other individuals; these are the relationships between individual members of the class and other items.



Step 6. Define the facets of the slots

- **Slot cardinality** defines how many values a slot can have.
- **Slot-value type**
 - what **types of values** can fill in the slot.
 - **common value types**: String, Number, Boolean, Enumerated, value type, allowed values, the number of the values (cardinality), other features of the values the slot can take.



Step 7 - rules slot Domain and Ranges

- When defining a domain or a range for a slot, **find the most general classes or class that can be respectively the domain or the range for the slots** . On the other hand, **do not** define a domain and range that is **overly general** (i.e., one would not want to make the range THING) but one would want to choose a class that will cover all fillers.



- If a list of classes defining a range or a domain of a slot includes a class and its subclass, remove the subclass.
- If a list of classes defining a range or a domain of a slot contains all subclasses of a class A, but not the class A itself, the range should contain only the class A and not the subclasses.
- If a list of classes defining a range or a domain of a slot contains all but a few subclasses of a class A, consider if the class A would make a more appropriate range definition.



Step 8 - Creating Instances

- Body: Light
- Color: Red
- Flavor: Delicate
- Tannin level: Low
- Grape: Gamay (instance of the Wine grape class)
- Maker: Chateau-Morgon (instance of the Winery class)
- Region: Beaujolais (instance of the Wine-Region class)
- Sugar: Dry



Step 9 - Defining classes and a class hierarchy

DIKE
Research
Group

- Ensuring that the class **hierarchy** is correct
- An “is-a” relation
- A **subclass** of a class represents a concept that is a “kind of” the **concept** that the superclass represents.
 - A single plant is not a subclass of all plants
- **Transitivity** of the hierarchical relations



- Evolution of a class hierarchy
 - distinction between classes and their names
 - hence synonyms of concept name do not represent different classes
 - Avoid class hierarchy cycles
 - All the siblings in the hierarchy (except for the ones at the root) must be at the same level of generality.



- How many is too many and how few are too few?
 - If a class **has only one** direct subclass there may be a modeling problem or the ontology is not complete.
 - If there are **more** than a dozen subclasses for a given class then additional intermediate categories may be necessary. (may not always be possible)



- **Multiple Inheritance**
- When do you introduce a new class
- **Subclasses of a class** usually
 - have **additional properties** that the superclass does not have, or
 - **restrictions different** from those of the superclass, or
 - **participate in different relationships** than the superclasses



- Counter example to thumb rule in previous slide
 - Classes in terminological hierarchies **do not have** to introduce new properties
 - MeSH at NLM National Institute of Health
 - purpose is just **to organize for ease of searching/indexing**



- **A new class or a property value?**
 - Class White Wine or simply property of class Wine that takes value White
 - **Depends on how important a concept** White Wine is in the domain
- **An instance or a class?**
 - starts with deciding **what is the lowest level of granularity in the representation.**
 - Individual instances are the most specific concepts represented in a knowledge base.



Step 10 - Limiting the scope

- The ontology **should not contain all the possible information about the domain**: you do not need to specialize (or generalize) **more than you need** for your application (at most one extra level each way).
- The ontology **should not contain all the possible properties** of and distinctions among classes in the hierarchy.



More details ...

- Inverse slots
- Naming conventions
- Synonyms
- Defaults
- Disjoint subclasses

Restricted to FATISDA



Protégé

- Plant.owl
- Ethno.owl
- Example.owl (expand from previous example)



Further Reading

- A Semantic Web Primer. The MIT Press Cambridge, Massachusetts, London, England. Grigoris Antoniou and Frank van Harmelen.
- <http://www.w3.org/TR/owl2-primer/>
- Ontology Development 101: A Guide to Creating Your First Ontology Natalya F. Noy and Deborah L. McGuinness Stanford University, Stanford, CA, 94305